Model Predictive Control
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Recitation 05
CFTOC

Exercise 1

Write the optimization problem

$$
\begin{aligned}
\min_{x \in \mathbb{R}^n} \quad & ||x||_p \\
\text{subj. to} \quad & Ax \leq b
\end{aligned}
$$

as Linear Program (LP) for

i) $p = 1$

ii) $p = \infty$

Exercise 2

Write the optimization

$$
\begin{aligned}
J^\star(x(k)) = \min_{x_i, u_i} \ & x_N^\top P x_N + \sum_{i=0}^{N-1} x_i^\top Q x_i + u_i^\top R u_i \\
s.t. \ & x_{i+1} = A x_i + B u_i \\
& G_x x_i \leq b_x, \qquad \forall i = 0, \ldots N-1 \\
& G_u u_i \leq b_u \\
& G_f x_N \leq b_f \\
& x_0 = x(k)
\end{aligned}
$$

as a QP with

$$
\begin{aligned}
J^\star(x(k)) = \min_{z} \ & z^\top H z + h^\top z + c \\
s.t. \ & G z \leq w \\
& G_{eq} z = E_{eq} x(k)
\end{aligned}
$$